

Session 5 Course Note: Evaluating, Investing, and Experimenting with AI

For students in Innovation and Technology Management

Workshop connection: Session 5, 13:30–14:00

Purpose: Integrate value, evidence, implementation, people, and risk into a disciplined AI investment decision.

How to use this note

The first decision is rarely whether to deploy AI across an organization. A more useful decision is whether there is enough value and control to justify a bounded experiment.

After reading, you should be able to:

- Evaluate an AI proposal as a business and organizational experiment.
- Distinguish activity, output, outcome, impact, and risk measures.
- Build a baseline and comparison.
- Estimate costs beyond the software licence.
- Design a limited, reversible pilot.
- Set continuation, change, and stop criteria.
- Present and assess a concise management recommendation.

1. Investment under uncertainty

AI proposals contain several uncertainties at once:

- **Technical:** Can the system perform the task reliably?
- **Data:** Are suitable and permitted data available?
- **Operational:** Can the system fit the workflow and existing systems?
- **Human:** Will users understand, adopt, challenge, and improve it?
- **Economic:** Will benefits exceed total costs?
- **Risk:** Are safety, security, rights, and reputation protected?
- **Strategic:** Does the capability support the organization's direction and resilience?

When uncertainty is high, committing to a large rollout creates risk and reduces the ability to learn. A small pilot can be viewed as purchasing information. Management pays a limited cost to discover whether the assumptions are credible.

The decision is therefore not:

Is AI good or bad?

It is:

What is the smallest investment that can produce decision-relevant evidence without exposing the organization to disproportionate harm?

2. What makes a proposal investable?

A credible first-stage proposal contains at least eight elements.

1. A defined problem and user

State who experiences the problem, when it occurs, how work happens today, and why the result matters.

2. A bounded AI role

Specify what the system may retrieve, draft, classify, predict, recommend, or do. State explicit exclusions.

3. A baseline

Measure present time, quality, cost, safety, delivery, or experience. If you do not know the baseline, make baseline measurement part of the plan.

4. A value hypothesis

Connect the AI role to one or two observable outcomes. Avoid unsupported financial precision.

5. A risk hypothesis

Identify the most credible harm, not only a generic risk list. Define how it will be detected.

6. A human operating model

Name the users, reviewer, approver, exception handler, incident owner, and accountable manager.

7. A smallest useful test

Limit task, data, users, location, period, integrations, and autonomy.

8. Decision rules

Define what evidence supports continuation, change, scale, or termination.

3. Separate activity from value

Use a measurement chain.

Level	Question	Example
Input	What resources were provided?	Licences, training hours, approved documents
Activity	What did people or the system do?	Queries, generated briefs, completed reviews
Output	What immediate work was produced?	Briefs completed per week
Quality	Was the output correct and usable?	Supported-claim rate, correction rate
Outcome	Did the workflow improve?	Reduced verified-search time, less rework
Impact	Did the wider objective improve and persist?	Lower downtime, better service, reduced cost
Guardrail	Was unacceptable harm avoided?	No unauthorized data access or safety error

High adoption is not proof of value. Low adoption is not automatically a user problem; the task may be unsuitable, the tool may be difficult to access, or employees may not trust the process.

Example: interpreting NBIM's 15% figure

NBIM's 2024 annual report says that employees reported an average 15% productivity improvement from AI tools. The report also describes strategy, model access, learning activities, and internal technology days (NBIM Annual Report 2024, p. 54 (https://www.nbim.no/contentassets/490f9f062cfc4694b12c45f4d04ab0a5/gpfg_annual_report_2024_uuweb2.pdf)).

The careful interpretation is:

- It is an organizationally relevant self-reported estimate.
- It indicates perceived usefulness and adoption experience.
- It is not a controlled causal estimate of output, quality, profit, or return on investment.

Evidence should be used at the level it supports.

4. Build a baseline and counterfactual

Baseline

The baseline describes current performance before the intervention. Use the same definition and measurement method during the pilot.

For a service-knowledge assistant:

- Median time from query to verified information.
- Percentage of cases using the current approved document.
- Number of escalations.
- Rework caused by incorrect information.
- User workload and satisfaction.

Counterfactual

The counterfactual asks what would have happened without the AI intervention. It matters because performance may improve for unrelated reasons such as training, seasonal workload, new staff, better data, or process redesign.

Possible comparisons include:

- Before and after for the same users.
- AI-assisted cases versus comparable current-practice cases.
- A phased introduction across teams.
- Random assignment where appropriate and ethical.
- Repeated measurement over time.

No design removes every uncertainty. State the limitations honestly.

5. What productivity studies can and cannot tell you

Research has found meaningful gains in some structured knowledge-work settings:

- A field study of a customer-support assistant found an average productivity increase of roughly 14–15%, with larger effects for less experienced workers and substantial variation across workers (Brynjolfsson, Li, and Raymond (<https://www.nber.org/papers/w31161>); later published in the *Quarterly Journal of Economics*).
- An experiment on professional writing tasks found that participants with ChatGPT completed work faster and received higher quality ratings on average (Noy and Zhang, 2023 (<https://pubmed.ncbi.nlm.nih.gov/37440646/>)).
- The FFI/Navy field experiment found that frequent users of a bounded HR assistant could answer recurring questions faster, while also identifying integration, trust, experience, and adoption issues (FFI report 25/008 (<https://www.ffi.no/publikasjoner/arkiv/okt-produktivitet-med-kunstig-intelligens-et-feltekspirement-med-digital-hr-assistent>)).

These studies do not provide a universal ROI figure. Effects depend on:

- Task structure.
- Model and interface.
- Data and knowledge sources.
- Worker experience.
- Training and workflow design.
- Output measurement.
- Whether the task lies within the system's capability.

The 2026 Stanford AI Index summarizes evidence that productivity gains tend to be larger in structured, measurable work and smaller in tasks requiring deeper reasoning. It also notes possible learning costs from heavy reliance (Stanford HAI, 2026—Economy (<https://hai.stanford.edu/ai-index/2026-ai-index-report/economy>)).

Use external studies to form a hypothesis, not to replace local evidence.

6. Measure quality and risk with value

A pilot that measures only speed encourages unsafe shortcuts.

For each value measure, add a quality or risk counterpart.

Value measure	Possible quality or risk counterpart
Time saved	Correction and rework time
Faster response	Supported-claim rate
More cases processed	Severity of missed cases
Higher adoption	Required verification completed
Lower inventory	Stockouts and service failures
Better prediction	False-negative rate by context or group
Automated action	Unauthorized or irreversible actions

Leading and lagging indicators

- **Leading indicators** show whether the process is being followed: training completed, source versions current, reviews performed.
- **Lagging indicators** show later results: incidents, complaints, rework, downtime, financial outcome.

You need both. A lack of reported incidents may mean the system is safe, or it may mean incidents are not detected or reported.

7. Calculate total cost, not licence cost

The total cost of ownership may include:

- Software licences or API use.
- Data cleaning, labelling, migration, and document governance.
- Integration with identity, workflow, and enterprise systems.
- Security, privacy, legal, procurement, and compliance work.
- Testing and independent assurance.
- User training and change management.
- Human review and exception handling.
- Monitoring, logging, incident response, and audits.
- Supplier management and model changes.
- Infrastructure, energy, and operational support.
- Exit, migration, and data-deletion costs.

Also consider **opportunity cost**: which other project, process improvement, or training investment is displaced?

Avoid false ROI precision

Early business cases often multiply optimistic time savings by salary without considering:

- Whether saved minutes can be converted into useful capacity.

- Whether review and rework move to another role.
- Whether demand for the task is fixed.
- Whether quality changes.
- Whether the improvement persists after novelty.
- Whether only enthusiastic volunteers were measured.

Use ranges and scenarios when evidence is weak.

8. Design a bounded pilot

A 60- or 90-day pilot can use the following structure.

Purpose

One decision-relevant question, for example:

Can a source-grounded assistant reduce time to verified service information without increasing unsupported technical claims?

Scope

- One task.
- A defined user group.
- A limited product, site, or document collection.
- A fixed period.
- Approved tools and integrations.

Exclusions

- Decisions the AI cannot make.
- Data it cannot access.
- People or cases outside the test.
- Actions requiring separate approval.

Measures

- Baseline and comparison.
- One or two value outcomes.
- At least one quality measure.
- At least one risk or guardrail measure.
- Adoption and process measures needed to interpret the result.

Roles

- Executive or business sponsor.
- Product or process owner.
- Technical owner.
- Data or source owner.
- Risk, privacy, security, or legal support as required.
- User representatives.
- Incident and stop authority.

Review points

- Readiness review before use.

- Early review after the first cases.
- Midpoint review.
- Final evidence review and decision.

Stop conditions

- Unauthorized data exposure.
- Untraceable safety-critical output.
- Error or disparity above threshold.
- Users cannot perform required verification.
- The system creates greater review or rework cost than the value achieved.

9. Use stage gates instead of one large commitment

An AI initiative can progress through gates.

Gate 0 — Problem and permission

Is the problem real, and is the proposed purpose permitted?

Gate 1 — Data and feasibility

Are the required data and technical capability sufficient for a controlled test?

Gate 2 — Pilot readiness

Are users, controls, measures, incident procedures, and stop authority ready?

Gate 3 — Pilot evidence

Did value appear without unacceptable harm, and are the limitations understood?

Gate 4 — Scale readiness

Can data governance, support, training, monitoring, supplier capacity, and accountability scale with use?

Gate 5 — Continued operation

Does the system remain useful, secure, compliant, and cost-effective after models, data, work, and regulation change?

Each gate can produce **continue**, **continue with conditions**, **redesign**, **pause**, or **stop**.

10. Portfolio thinking

Organizations usually have more AI ideas than capacity to test them. Compare proposals on a common set of questions:

Dimension	Strong early-pilot signal	Warning signal
Problem	Frequent, costly, and well understood	Vague innovation ambition
Task	Bounded and measurable	Broad, shifting, or subjective
Data	Permitted, relevant, current, traceable	Fragmented, restricted, or unknown
Value	Meaningful outcome and baseline	Only usage or speculative ROI
Consequence	Low to moderate and reversible	High impact and difficult to reverse
Human control	Named, competent, and resourced	Generic “human in loop”
Testability	Small comparison possible	Value only visible after full rollout
Exit	Can stop and migrate	Supplier or integration lock-in

Do not automatically fund the proposal with the highest imagined upside. A narrower proposal may produce better evidence and strategic learning.

11. Decide what “scale” means

Scaling may involve:

- More users.
- More locations.
- More languages.
- More products or datasets.
- More integrations.
- More autonomy.
- Higher-consequence decisions.

Each is a different change. Evidence from one does not automatically justify the others.

For example, a system that assists ten engineers in English may require new testing before it is used across multiple languages, connected to customer data, or allowed to create work orders.

Scale controls with value:

- Document and data governance.
- User training and support.
- Monitoring and incident capacity.
- Access management.
- Fairness and performance testing.
- Supplier assurance.
- Human review resources.

12. The 90-second management pitch

Use four parts.

1. Problem and AI role

Who needs help? What will the system do, and what is excluded?

2. Value and evidence

Which outcome should improve? What baseline or existing evidence makes that plausible?

3. Risk and safeguard

What is the most credible harm? Which safeguard is non-negotiable, and who owns it?

4. Test and decision request

What is the smallest useful test? Are you asking for go, conditional go, or redesign?

Example:

Field-service engineers lose time finding the current procedure for unfamiliar alarms. We propose a six-week assist-only pilot that retrieves and summarizes approved documents for two equipment families; it will not diagnose or create work orders. We will compare median verified-search time and correct-version use with current practice. The main risk is an unsupported safety statement, so every claim must link to an approved passage and the service lead can stop the test immediately. We request a conditional go after document permissions and baseline measurement are complete.

13. How to listen to a pitch

Do not confuse confidence, technical vocabulary, or presentation quality with proposal quality.

Listen for:

- A real user and problem.
- A bounded system role.
- A baseline and measurable outcome.
- Honest uncertainty.
- A credible harm and concrete safeguard.
- A named accountable owner.
- A small, reversible test.
- Evidence that would change the team's mind.

Your workshop investment card asks you to allocate NOK 1 million across no more than two proposals, or invest nothing. Your reason should connect value, learning, and control.

14. Career connection for ITM students

Organizations need more than AI developers. They need people who can connect:

- Industrial and domain knowledge.
- Technology and data.
- Strategy and finance.
- Workflows and operations.
- Organizational change and skills.
- Security, ethics, law, and governance.
- Experiment design and evidence.

This translation and integration role is central to Innovation and Technology Management. A strong manager does not need to know every model architecture, but must know which questions require technical, legal, operational, or domain expertise and how to bring those perspectives into one decision.

15. A post-workshop pilot checklist

Before funding a pilot, confirm:

- ☐ A named user and problem.
- ☐ A measured or planned baseline.
- ☐ A bounded AI role and explicit exclusions.
- ☐ Permitted and governed data or sources.
- ☐ A technical and non-AI alternative considered.
- ☐ Value, quality, and guardrail measures.
- ☐ A realistic comparison.
- ☐ Named owners and effective human review.
- ☐ Training, support, and incident procedures.
- ☐ Stop conditions and exit plan.
- ☐ A decision date and evidence threshold for continuation.

16. Reflection questions

1. Which measure in your proposal is activity, and which is an outcome?

2. What would have happened without your proposed AI system, and how will you estimate that?
3. Which cost is most likely to be omitted from your business case?
4. What evidence would justify wider use but not greater autonomy?
5. Under what result would stopping the pilot demonstrate good management rather than failure?

Key terms

- **Counterfactual:** An estimate of what would have happened without the intervention.
- **Outcome:** A change in the workflow, user, customer, or operation.
- **Impact:** A sustained wider effect attributable to the intervention.
- **Guardrail:** A boundary or measure designed to prevent or detect unacceptable harm.
- **Total cost of ownership:** All acquisition, implementation, operation, governance, and exit costs.
- **Stage gate:** A decision point requiring defined evidence before further commitment.
- **Pilot:** A limited real-world test designed to generate evidence.
- **Scale:** Expansion in users, contexts, data, integrations, autonomy, or consequence.
- **Exit plan:** The technical and organizational method for stopping use and removing dependencies safely.

References and further reading

- Stanford Institute for Human-Centered AI. 2026 AI Index—Economy (<https://hai.stanford.edu/ai-index/2026-ai-index-report/economy>).
- Brynjolfsson, E., Li, D., and Raymond, L. Generative AI at Work (<https://www.nber.org/papers/w31161>), NBER Working Paper 31161; later published in the *Quarterly Journal of Economics*.
- Noy, S., and Zhang, W. Experimental evidence on the productivity effects of generative artificial intelligence (<https://pubmed.ncbi.nlm.nih.gov/37440646/>), *Science* 381 (2023).
- FFI. Økt produktivitet med kunstig intelligens? (<https://www.ffi.no/publikasjoner/arkiv/okt-produktivitet-med-kunstig-intelligens-et-felteksperiment-med-digital-hr-assistent>), report 25/008 (2025).
- NBIM. Government Pension Fund Global Annual Report 2024 (https://www.nbim.no/contentassets/490f9f062cfc4694b12c45f4d04ab0a5/gpfg_annual_report_2024_uuweb2.pdf), especially p. 54.
- NIST. AI Risk Management Framework (<https://www.nist.gov/itl/ai-risk-management-framework>) and AI Resource Center (<https://airc.nist.gov/>).

Session takeaway

Invest first in a bounded learning cycle. Scale only when the evidence, operating model, and controls can scale with the value.